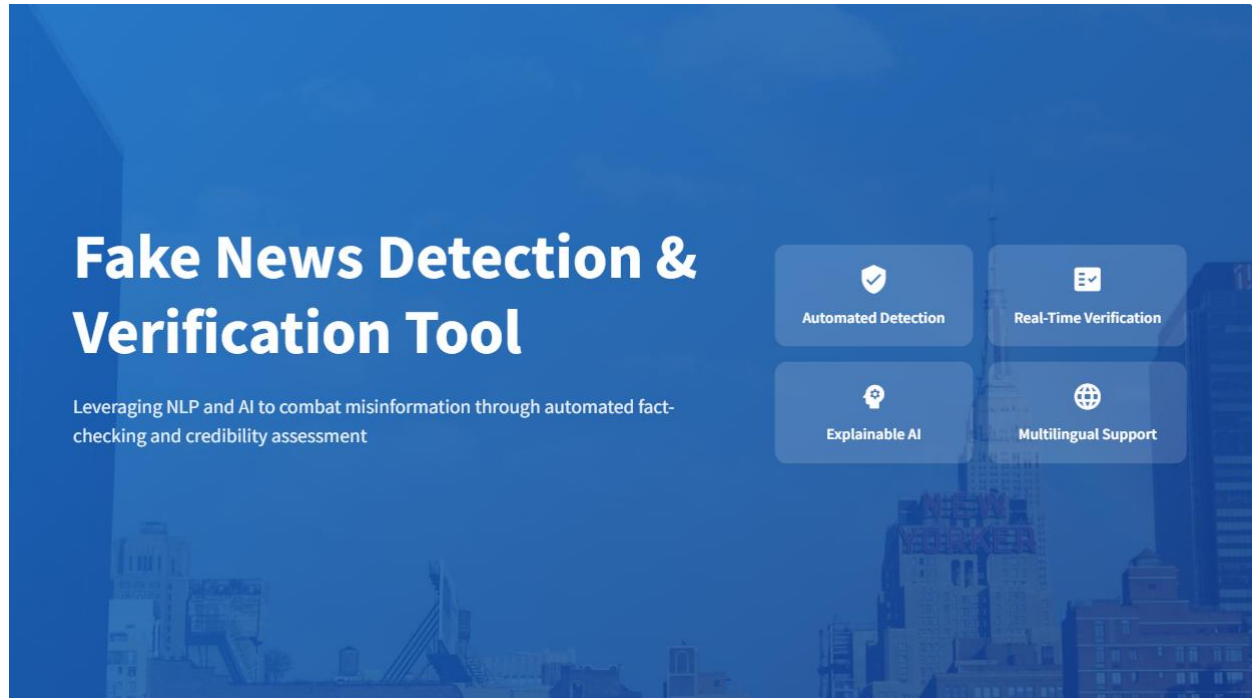


Fake News Detection & Verification Tool



2. Project Statement

In today's digital world, the rapid spread of misinformation through news articles and social media posts poses significant risks to public trust, political stability, and public safety. Manual fact-checking is slow and resource-intensive, making it difficult to address false information in real time.

The **Fake News Detection & Verification Tool** aims to combat misinformation by leveraging **Natural Language Processing (NLP)**, **deep learning models**, and **real-time fact-checking APIs**. The system will analyze textual content, assess its credibility, and cross-reference facts with reliable sources. Additionally, it will provide explainable results, highlight suspicious claims and linking users to verified references.

This open-source project will be designed to **engage 25 interns over 8 weeks**, with clearly defined modules covering preprocessing, classification, claim verification, and visualization, ensuring collaboration while applying the latest AI advancements in misinformation detection.

3. Outcomes

- **Automated Fake News Detection:** Classify content as reliable, suspicious, or fake using deep learning.

- **Real-Time Claim Verification:** Cross-check extracted claims with trusted fact-checking APIs (e.g., Google Fact Check Tools, PolitiFact).
- **Explainable AI Insights:** Highlight key phrases and claims that influenced the classification.
- **Source Credibility Scoring:** Assign credibility ratings to publishers based on historical accuracy.
- **Multilingual Support:** Detect misinformation in multiple languages using transformer-based models.
- **User-Friendly Interface:** Intuitive web platform for news/article input and result visualization.

4. Modules to be Implemented

1. User Authentication & Profile Management

- Registration/login system with JWT authentication.
- User profile for managing submitted articles and analysis history.

2. News Article & Social Media Post Ingestion Module

- Paste or upload text, or fetch from a URL/social media link.
- Text extraction and cleaning (handling hashtags, emojis, metadata).

3. NLP Preprocessing & Feature Extraction Module

- Tokenization, stopword removal, lemmatization (NLTK/spaCy).
- Named Entity Recognition to identify people, places, and organizations.

4. Fake News Classification Module

- Transformer-based models (e.g., BERT, RoBERTa) fine-tuned on fake news datasets.
- Output: Probability score (0–1) and classification (real/suspicious/fake).

5. Claim Verification Module

- Extract factual claims using dependency parsing and NER.
- Verify against trusted fact-checking APIs and databases.

6. Explainability & Visualization Module

- Highlight suspicious phrases in text.
- Display verification results with supporting source links.

7. Admin Dashboard

- Manage trusted source list and API keys.
- Monitor system performance and usage statistics.

5. Week-Wise Milestones

Milestone 1 (Weeks 1–2): User & Input Management

- User authentication (JWT).
- Input module for articles/social media links.
- Basic text cleaning and preprocessing pipeline.

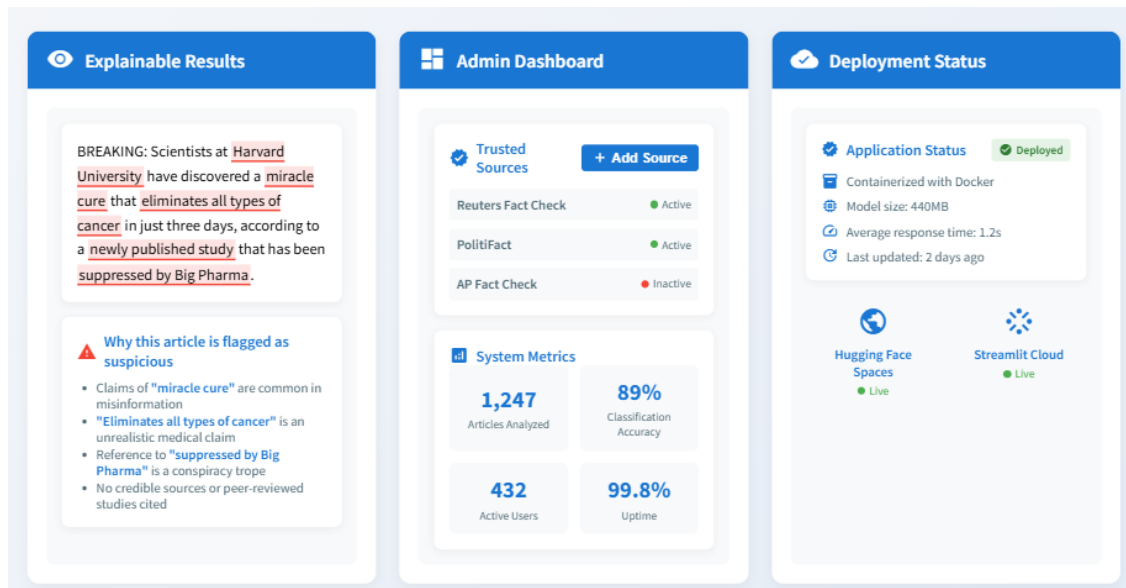
The image displays a three-panel user interface for a text analysis application. The first panel, 'User Authentication', contains fields for email (user@example.com) and password (masked with dots), and a 'Sign In' button. The second panel, 'Article Input Interface', offers three input methods: 'Paste or upload text', 'Enter URL or social media link', and 'Upload document file'. The third panel, 'Text Preprocessing', shows the 'Original Text' as a news snippet about a cure, with the word 'AMAZING' highlighted in blue. Below it, the 'Processed Text' shows the same snippet with 'AMAZING' removed and the text lowercased.

Milestone 2 (Weeks 3–4): Core NLP Processing

- Implement tokenization, lemmatization, and entity extraction.
- Store processed text and extracted entities in database.

Milestone 4 (Weeks 7–8): Explainability, Dashboard & Deployment

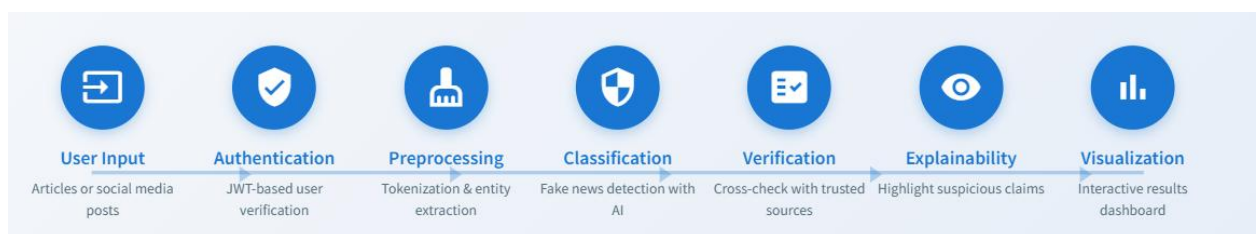
- Implement explainability (highlight suspicious claims).
- Build admin dashboard for source management.
- Containerize with Docker and deploy on Hugging Face Spaces/Streamlit Cloud.



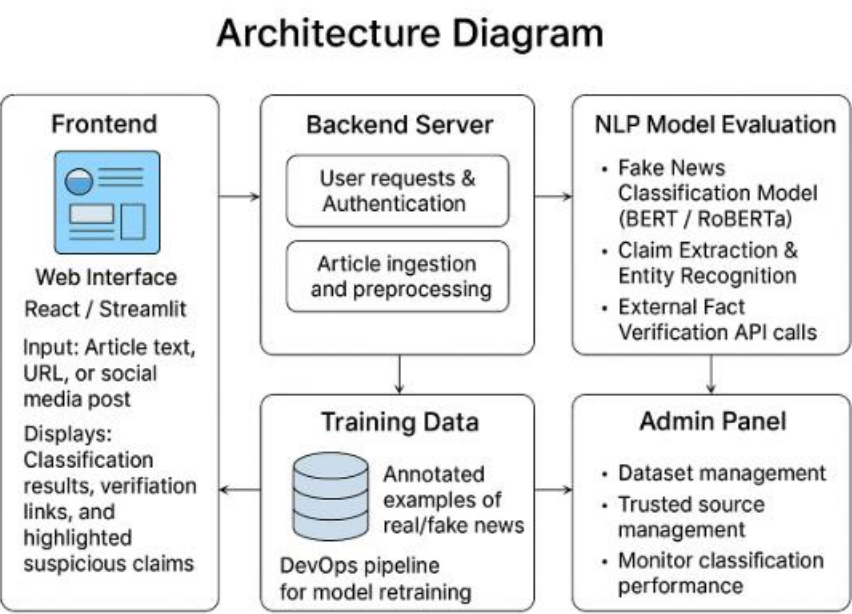
6. Evaluation Criteria

- **End of Week 2:** Functional login and text ingestion.
- **End of Week 4:** Working NLP pipeline with entity extraction.
- **End of Week 6:** Functional classification model with verification integration.
- **End of Week 8:** Explainable results, dashboard ready, deployed system, and complete documentation.

Work Flow



Architecture Diagram



Schema Diagram

